1. Clinical Question & Core Hypothesis

Clinical Question

Why do tumors with abundant stroma (high stromal-to-tumor ratio) not always show better containment of disease, and under what molecular conditions does a fibrotic TME drive early progression? 2) Tier 1 paths used: (a) TGFB1/TGF-β fibrogenic signaling (Path 1) — gives mechanistic basis for stromal expansion and immune modulation; (b) Immune contexture (Immunoscore/CD8) (Path 7) — provides the anti-tumor effector axis whose exclusion could explain worse outcomes. I built on these because they directly link stromal biology to immune exclusion and clinical outcome. 3) Tier 2 support: GREM1 (metastasis/EMT), SMAD4 (downstream mediator of TGF-β transcriptional programs), EPCAM/CTC concepts (dissemination readouts) reinforce a pathway in which TGF-β–driven stroma promotes invasion/dissemination and immune evasion; AI/radiomic approaches provide ways to measure TSR and integrate multimodal signals. 4) Endpoint choice: PFI was chosen because the mechanistic model implicates stromal-driven immune exclusion and dissemination as drivers of disease progression (PFI captures time to progression), and PFI is robustly available in TCGA-BRCA-like datasets for testing progression hypotheses. 5) Deliberate exclusions: I excluded imaging features focused on lymphoid aggregates/TIL density as the primary imaging variable to keep the hypothesis sharply focused on stromal/fibrotic mechanisms; I also excluded broad multi-gene signature claims to satisfy the 1–2 anchor constraint and to emphasize an explicit mechanism (TGF-β–mediated immune exclusion) rather than an associative predictor?

Working Hypothesis

In bladder cancer, high stromal-to-tumor ratio reflects TGF-β–driven fibrotic remodeling (via TGFB1 signaling) that promotes an immune-excluded microenvironment. Consequently, tumors with high TSR and elevated TGFB1 but low CD8A expression will have a shortened progression-free interval (PFI) compared with tumors lacking this triad, highlighting a mechanism where fibrosis-mediated immune exclusion accelerates progression.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TGFB1, CD8A
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGFB1: TGFB1/TGF-β fibrogenic signaling (Path 1) — gives mechanistic basis for stromal expansion and immune modulation; (b) Immune contexture (Immunoscore/CD8) (Path 7) — provides the anti-tumor effector axis whose exclusion could explain worse outcomes.

CD8A: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	GREM1	Supporting modifier from knowledge graph
Secondary	SMAD4	Supporting modifier from knowledge graph
Secondary	EPCAM	Supporting modifier from knowledge graph
Secondary	CTCs	Supporting modifier from knowledge graph
Secondary	Immunoscore	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: induces fibroblast activation and ECM deposition, expanding fibrotic stroma that correlates with higher TSR. This fibrotic milieu can create physical and biochemical barriers that limit CD8+ T cell infiltration, reflected by lower CD8A expression, thereby enabling tumor progression and potential resistance to immune therapies.

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- GREM1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- SMAD4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.